

Homework 1

Due Tuesday October 13th at 11:59 PM PT

This homework is worth 30 points and covers Lectures 1–4 and Labs 1–2. Write your solutions in an R Markdown file and submit the PDF to Gradescope. Show all code used in your solutions. Unless stated otherwise, use $\alpha = 0.05$.

Introduction

Suppose you are a data scientist on a university teaching-and-learning research team. You are analyzing a 2025 survey of 357 undergraduates at a public university in Colombia about their study habits and AI use. Students from 14 academic programs were included, with larger programs contributing more responses. All responses are anonymized.

The tab-separated file `ai_study_habits.tsv` contains one row per student. You will use three variables:

- `ai_use_frequency`: how often the student reports using AI tools for studying;
- `ai_tools`: a semicolon-separated list of AI tools reported by the student; and
- `gpa`: the student’s self-reported GPA on a 0–5 scale.

For `ai_use_frequency`, responses of “Almost always” or “Always” identify a *frequent AI user*.

For the hypothesis-test calculations, treat the rows as independent observations from the same population. In Question 2(c), you will discuss whether this assumption is reasonable. The survey is cross-sectional and self-reported, so the results do not establish causation.

Question 1 [15 points]

The research team wants to know whether frequent AI use is common enough to motivate campus-wide AI guidance. Let p be the true proportion of undergraduates at this university campus who are frequent AI users for studying. We test

$$H_0 : p = 0.50 \quad \text{versus} \quad H_A : p > 0.50.$$

The alternative asks whether more than half of students are frequent AI users. For a hypothetical pilot analysis, suppose only 9 responses are available. We will assess whether this small sample can detect a meaningful majority and whether more data are needed.

(a) [4 points] You would like to use $\alpha = 0.05$, but because the binomial distribution is discrete, an exact significance level of 0.05 is not attainable here. Write code to show that the upper-tail rejection region with significance level closest to 0.05 (either above or below it) is $\{8, 9\}$. State the actual significance level and briefly explain how you found it and what it means in context.

Hint: The `pbinom` function may be useful. Pay careful attention to the `lower.tail` argument, strict versus non-strict inequalities, and the fact that a p-value includes results at least as extreme as the observed result.

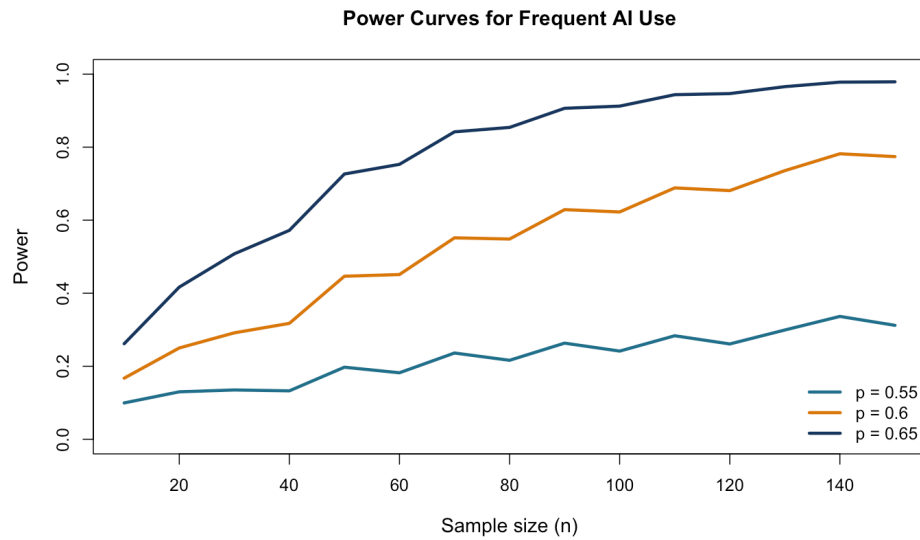
(b) [2 points] Using the rejection region from part (a), suppose the true proportion of frequent AI users is $p = 0.65$. Find the power of the test. Explain what this value means and why you consider the power high or low.

(c) [2 points] Suppose 7 of the 9 students in the pilot sample are frequent AI users. Using the rejection rule from part (a), state the test decision and compute the exact one-sided p-value. Briefly explain why 7/9, although well above 0.50, does not lead to rejection.

(d) [4 points] Write code to recreate the supplied power curves for true proportions $p = 0.55, 0.60$, and 0.65 , and sample sizes from 10 to 150 in increments of 10. For each sample size, use the upper-tail rejection region whose exact significance level is closest to 0.05. You may use the following cutoffs:

```
rej_regions_cutoff <- c(8, 14, 20, 26, 31, 37, 42, 48,  
                        53, 59, 64, 70, 75, 80, 86)
```

Include the plot and summarize what the curves show.



(e) [3 points] Load `ai_study_habits.tsv`. Create a logical or numeric variable identifying frequent AI users, and use all 357 observations to perform the hypothesis test stated at the beginning of Question 1. Report:

Hint: `read.delim()` may be useful for loading a tab-separated file.

- the number and sample proportion of frequent AI users;
- the exact one-sided p-value; and
- a brief conclusion in context.

Reminder: all code must be shown for full credit.

Question 2 [7 points]

The research team also wants to know whether students typically use multiple AI tools, rather than one tool or none. This may help determine whether student guidance should focus on a single platform or on using and evaluating several tools.

The variable `ai_tools` lists the tools reported by each student, separated by semicolons. For example,

`ChatGPT; DeepSeek; Copilot; Gemini`

means that the student reported four tools. A blank value means that the student reported no tools. Create a numeric variable `n_ai_tools` containing the number of tools reported by each student. You will use this variable throughout Question 2.

Hint: `strsplit(x, ";")` splits a character value at each semicolon. You may find `length()` and `sapply()` useful. Be careful with blank values.

(a) [1 point] Plot a histogram of `n_ai_tools`. Use a meaningful title and axis labels.

(b) [4 points] We are interested in the population median number of reported AI tools, $\tilde{\mu}$. The median is a useful measure of center for this right-skewed distribution. Because the tool count is integer-valued, 1.5 separates one or fewer tools from two or more tools. Use a sign test to test

$$H_0 : \tilde{\mu} = 1.5 \quad \text{versus} \quad H_A : \tilde{\mu} > 1.5.$$

Find the upper-tail rejection region whose exact significance level is closest to 0.05. State the rejection region and its actual significance level. Then report the exact p-value and your conclusion.

(c) [2 points] The sign test assumes independent observations. The students came from one public-university campus in Colombia. Students from all academic programs were included, with larger programs contributing more students, and quotas were used within each program rather than taking a simple random sample of all university students. Briefly discuss:

1. whether independence is reasonable; and
2. how broadly the results can be generalized.

A well-justified argument either for or against independence will receive full credit.

Question 3 [8 points]

For this hypothetical analysis, suppose the university chose 3.9 in advance as a reference GPA for academic reporting. The research team wants to know whether the students represented in this survey are centered differently from this value.

(a) [1 point] Plot a histogram of `gpa`. Use a meaningful title and axis labels.

(b) [3 points] The GPA distribution is fairly symmetric and bell-shaped, so a one-sample t-test is reasonable for the population mean. We will also use a sign test for the population median. Perform a t-test for

$$H_0 : \mu = 3.9 \quad \text{versus} \quad H_A : \mu \neq 3.9,$$

and a sign test for

$$H_0 : \tilde{\mu} = 3.9 \quad \text{versus} \quad H_A : \tilde{\mu} \neq 3.9,$$

where μ and $\tilde{\mu}$ are the population mean and median GPA, respectively. For the sign test, remove observations exactly equal to 3.9 before counting signs. Report the p-value and conclusion from each test, and comment briefly on why their p-values need not be identical.

(c) [4 points] To choose a method for future monitoring, the team also wants to compare statistical power. When its assumptions hold, the t-test can be more powerful for detecting a change in the mean.

Write a simulation showing at least one normally distributed scenario in which a two-sided t-test has at least 15 percentage points more estimated power than a two-sided sign test, using the same null value. You may use a p-value cutoff of 0.05 for both tests, even though the sign test's exact significance level may differ from 0.05. You may need to try several parameter combinations. If a scenario does not meet the target, change the model parameters rather than repeatedly changing the random seed.

Use at least 5,000 repetitions. Set and report a random seed. Report:

- the null value, true mean, standard deviation, and sample size;
- the estimated power of each test; and
- a brief comparison of the two estimates.